

AMANDEEP SINGH

Software Developer | Node.js & AI Integration

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SUMMARY

MCA fresher specializing in backend development with hands-on experience building production-grade systems. Built ShodhKartaChecker (AI-powered plagiarism detection engine) using Node.js, FastAPI, Redis task queues, and Groq AI, achieving 40% faster API response times.

Skilled in microservices architecture, JWT authentication, RBAC, and REST API design. Looking to contribute to a backend-focused team and solve real-world engineering problems.

SKILLS

- Languages: JavaScript (Node.js), Python, Typescript
- Frameworks: Express.js, React.js, Nest.js
- Databases: MySQL, MongoDB, Redis, PostgreSQL
- ORMs: Prisma , Mongoose , TypeORM
- Tools: Docker, Git, GitHub, Postman, Nginx, PM2
- Cloud & Storage: Cloudinary , MongoDB Atlas
- Deployment: VPS, Vercel, Render
- Technologies: REST APIs, JWT Authentication, API Security, AI API Integration
- Concepts: Data Structures, DB Design, Error Handling, MVC Architecture, Backend Optimization, Asynchronous Processing

CERTIFICATION AND TRAINING

Full Stack Web Development Training – Department of Computer Science Punjabi University, Patiala

(MCA Industrial Training Program)

Duration: 1 Jan 2026 – 5 May 2026

- Learned and worked JavaScript, React.js, Node.js, Express.js, MongoDB, Python and Redis
- Developed backend applications and AI-integrated projects during training
- Built an AI-Powered Plagiarism Detection System with Redis-based task processing and Groq AI API integration

PROJECTS

ShodhKartaChecker (AI-Powered Plagiarism Detection System) [GitHub](#) | [Live Demo](#)

- Built a distributed plagiarism detection platform using Node.js, Express.js, FastAPI, Redis, and MongoDB Atlas
- Designed a decoupled microservices architecture where Node.js handled API orchestration and FastAPI processed heavy document analysis workloads

- Implemented Redis-based asynchronous task queues to improve API responsiveness and background document preprocessing
- Integrated Groq AI APIs for AI-generated plagiarism explanations and similarity analysis
- Configured Nginx reverse proxy and PM2 process management for production deployment on VPS
- Optimized document-processing workflows, reducing API response time by 40%

Schedula (Doctor appointment Booking System) [GitHub](#) | [Live Demo](#)

- Built a production-grade appointment booking REST API using NestJS and TypeScript with role-based access for Doctors and Patients
- Implemented JWT authentication and RBAC — doctors can manage availability and time slots, patients can search and book appointments
- Designed PostgreSQL database schema with TypeORM for managing users, appointments, and doctor schedules
- Built doctor search with filters and pagination for efficient data retrieval
- Documented all APIs using Swagger UI for easy testing and integration reference
- Deployed on VPS with Docker and Nginx reverse proxy for production reliability

PostGrid (Blog Platform) [GitHub](#) | [Live Demo](#)

- Built a RESTful blog platform using Node.js and Express.js with full CRUD operations — users can create, edit, delete posts and view a live feed of others' content
- Implemented JWT-based authentication and secure user profile management
- Integrated Cloudinary for cloud-based media storage and Multer for multipart image upload handling
- Used Prisma ORM with PostgreSQL for type-safe database queries and schema management
- Containerized all services with Docker and deployed on VPS for production hosting

EDUCATION

Master of Computer Applications (MCA)
 Punjabi University – Patiala, India
 Expected: May 2026